ALTRODAV TECHNOLOGIES • TECHNICAL ASSESSMENT

AI/ML Developer: Task 8

Topic: Machine Learning Model Development & **Difficulty:** Intermediate **Duration:** 4 Hours

1. PROBLEM STATEMENT

A learning platform wants to predict whether a student is likely to successfully complete a training program based on historical learning behavior and assessment performance.

Your objective is to build and evaluate machine learning models using a structured dataset. The goal is to identify important factors influencing student success and provide actionable insights from the model results.

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2. BUSINESS QUESTIONS TO ANSWER

Q1:

Which features have the highest impact on predicting student success?

Q2:

Can student completion status be predicted accurately using learning activity data?

Q3:

Which machine learning algorithm performs best for this classification problem?

Q4:

How do attendance, assessment scores, and activity levels influence predictions?

Q5:

Which students are predicted to be at risk of not completing the program?

Q6:

What is the precision, recall, and F1-score of each model?

Q7:

How can the model's predictions help improve learning outcomes?

Q8:

What recommendations can be made based on model findings?

3. TECHNICAL REQUIREMENTS

Dataset Preparation

Use or create a dataset containing:

- Student ID
- Attendance Percentage
- Assessment Score
- Assignment Completion Rate
- Login Frequency
- Study Hours
- Activity Count
- Final Completion Status (Target Variable)

Data Preprocessing

Perform:

- Missing value handling
- Duplicate removal
- Feature encoding
- Feature scaling
- Exploratory Data Analysis (EDA)

Model Development

Train at least **3 Machine Learning Models**:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Model Evaluation

Evaluate using:

- Accuracy Score
- Precision
- Recall

- F1 Score
- Confusion Matrix

Visualization Requirements

Create:

- Correlation Heatmap
- Feature Importance Chart
- Model Performance Comparison Chart
- Confusion Matrix Visualization

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4. SUBMISSION DELIVERABLES

ml_model.ipynb

Complete Jupyter Notebook containing:

- Data Cleaning
- EDA
- Feature Engineering
- Model Training
- Model Evaluation

dataset.csv

Dataset used for training and testing.

visualizations/

Folder containing all generated charts and visualizations.

REPORT.md

Document containing:

- Problem Statement
- Data Summary
- Model Comparison
- Key Findings
- Business Recommendations

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5. EVALUATION RUBRIC

Assessment Criteria	Weight
Data Preprocessing	20%
Model Development	25%
Model Evaluation	25%
Visualization & Analysis	20%
Documentation	10%

Excellent Standard

- Clean and well-structured ML pipeline.
- Proper feature engineering.
- Multiple models compared effectively.
- Strong interpretation of results.
- Professional documentation and visualizations.

Good Standard

- Models train successfully.
- Basic evaluation metrics included.
- Limited analysis of model performance.
- Minimal business insights.

- **Learning Outcome**

By completing this task, interns will gain practical experience in:

- Data Preprocessing
- Classification Algorithms
- Model Evaluation Techniques
- Feature Importance Analysis
- Machine Learning Workflow

- Business-Oriented AI/ML Reporting